

Supplementary Appendix: Semiclassical Spacetime Reshaping via Quantum Vacuum Stress-Energy Tensor at the Event Horizon

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1 Introduction and Physical Context

In standard classical General Relativity, the event horizon of a Schwarzschild black hole is governed entirely by the classical metric geometry dictated by the central mass. However, by incorporating quantum electrodynamics (QED) decoupling at the boundary ($r \rightarrow r_s$) where material electron charge density vanishes ($N_e \rightarrow 0$), the electromagnetic interaction term vanishes ($\mathcal{L}_{\text{int}} = 0$). Consequently, the physical state of the field transitions from a classically interacting field to a pure quantum vacuum state. To rigorously describe how this quantum vacuum behavior directly feeds back into and reshapes the spacetime geometry at the boundary, we transition from classical Einstein equations to the framework of Semiclassical Gravity.

2 The Semiclassical Einstein Field Equations

Under this paradigm, the geometry of spacetime represented by the Einstein tensor $G_{\mu\nu}$ is no longer driven by classical matter currents, but rather by the renormalized vacuum expectation value of the quantum energy-momentum tensor $\langle 0 | \hat{T}_{\mu\nu} | 0 \rangle_{\text{Renorm}}$.

The master equation governing this quantum-geometric transition at the event horizon is expressed as:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \langle 0 | \hat{T}_{\mu\nu} | 0 \rangle_{\text{Renorm}} \quad (1)$$

3 Detailed Variable Explanations

where the components of this equation are defined as follows:

- $G_{\mu\nu}$: The Einstein tensor, describing the geometric curvature and causal structure of the spacetime manifold.
- $\frac{8\pi G}{c^4}$: The coupling constant linking geometry to the source of energy and momentum in General Relativity.
- $\langle 0|\hat{T}_{\mu\nu}|0\rangle_{\text{Renorm}}$: The renormalized vacuum expectation value of the quantum energy-momentum tensor operator, representing pure quantum vacuum fluctuations and stress (such as Boulware or Hartle-Hawking vacuum states) arising at the boundary due to QED decoupling.

4 Physical Implications for Event Horizon Darkness

Equation (1) provides the mathematical bridge supporting the hypothesis that event horizon darkness is not a mechanical trapping effect caused by a classical gravitational pulling force. Instead, as the material charge current vanishes, the quantum vacuum stress acts as the active driver. This demonstrates that the boundary conditions are dictated by quantum vacuum dynamics, offering a refined perspective on how quantum states interact with spacetime geometry at the horizon.